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EDUCATION

- **Zhejiang University** Aug. 2019 - July, 2024
Ph.D. in College of Computer Science and Technology
Advisor: Prof. Jinsong Han Hangzhou, China
- **Wuhan University** Sep. 2015 - June 2019
B.Eng. in School of Computer
Advisor: Prof. Qian Wang Wuhan, China

RESEARCH INTERESTS

- Internet of Things Security (IoT Security)
- Mobile Computing
- Wireless Sensing
- Millimeter Wave

PUBLICATION

- **[Usenix Security'25] DiskSpy: Exploring a Long-Range Covert Channel via mmWave Sensing of μm -level HDD Vibrations.**
Weiye Xu, Danli Wen, Jianwei Liu, Zixin Lin, Yuanqing Zheng, Jinsong Han. [USENIX Security Symposium, 2025. (CCF-A)]
This paper reveals the risk of a first-of-its-kind covert channel, DiskSpy, which leverages mmWave signals to sense modulated HDD vibrations for leaking information from air-gapped computers. To push the limits of mmWave sensing in both precision and range, DiskSpy introduces a Tx-Rx co-optimization method, achieving the first μm -level vibration sensing at a long range of 22 meters using COTS mmWave radar.
- **[MOBICOM '22] Mask Does Not Matter: Anti-Spoofing Face Authentication using mmWave Without On-site Registration.**
Weiye Xu, Wenfan Song, Jianwei Liu, Yajie Liu, Xin Cui, Yuanqing Zheng, Jinsong Han, Xinhui Wang, Kui Ren. [Proceedings of the 28th Annual International Conference on Mobile Computing And Networking, 2022. (CCF-A, Acceptance rate: 17.8%)]
This paper exploits the penetrability, material sensitivity, and fine-grained sensing capability of millimeter wave (mmWave) to build an anti-spoofing FA system, mmFace. Additionally, to avoid inconvenient on-site registration, mmFace also proposes a novel virtual registration approach based on the core idea of cross-modal transformation from photos to mmWave signals.
- **[INFOCOM '21] RFace: Anti-Spoofing Facial Authentication Using COTS RFID.**
Weiye Xu, Jianwei Liu, Shimin Zhang, Yuanqing Zheng, Feng Lin, Jinsong Han, Fu Xiao, Kui Ren. [IEEE International Conference on Computer Communications, 2021. (CCF-A, **Best Paper Candidate, 7/251**)]
This paper proposes a novel privacy-preserving anti-spoofing FA system, named RFace, which extracts both the 3D geometry and inner biomaterial features of faces using a COTS RFID tag array. RFace builds a theoretical model to rigorously prove the feasibility of feature acquisition and the correlation between the facial features and RF signals.
- **[TMC '23] Anti-Spoofing Facial Authentication Based on COTS RFID.**
Weiye Xu, Jianwei Liu, Shimin Zhang, Yuanqing Zheng, Feng Lin, Fu Xiao, Jinsong Han. [IEEE Transactions on Mobile Computing, 2023. (CCF-A, Fast Tracked to IEEE TMC as the Best Paper Candidate in IEEE INFOCOM 2021)]
This paper designs a replay attack defense method which specifies the reading order of tags for each authentication to increase the difficulty of conducting replay attacks. Furthermore, our work designs a novel algorithm to enhance the flexibility by mitigating the impact of distance and deflection variations between the human face and tag array.

- **[CVPR '25] Your Scale Factors are My Weapon: Targeted Bit-Flip Attacks on Vision Transformers via Scale Factor Manipulation.**
Jialai Wang, Yuxiao Wu, Weiye Xu, Yating Huang, Chao Zhang, Zongpeng Li, Mingwei Xu, Zhenkai Liang.
[The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025. (CCF-A)]
- **[IWQoS '24] Manipulating Semantic Communication by Adding Adversarial Perturbations to Wireless Channel.**
Jianwei Liu, Yinghui He, Weiye Xu, Yifan Xie, Jinsong Han [IEEE/ACM International Symposium on Quality of Service, 2024.(CCF-B)]
- **[MOBICOM '20] Deaf-Aid: Mobile IoT Communication Exploiting Stealthy Speaker-to-Gyroscope Channel.**
Ming Gao, Feng Lin, Weiye Xu, Muertikepu Nuermainaiti, Jinsong Han, Wen Yao Xu, Kui Ren.
[Proceedings of International Conference on Mobile Computing And Networking, 2020.(CCF-A)]
- **[TON '23] Mobile Communication Among COTS IoT Devices via a Resonant Gyroscope with Ultrasound.**
Feng Lin, Ming Gao, Lingfeng Zhang, Yimin Li, Weiye Xu, Jinsong Han, Xian Xu, Wen Yao Xu, Kui Ren.
[IEEE/ACM Transactions on Networking, 2022. (CCF-A)]
- **[CCCF '23] A Discourse on Fine-Grained Wireless Sensing.**
Jinsong Han, Weiye Xu, Jianwei Liu, Wenfan Song. [Communication of the CCF, 2023]

UNDER REVIEW (FIRST AUTHOR)

- **Anti-Spoofing and Mask-Supported Face Authentication using mmWave without On-site Registration**
IEEE Transactions on Mobile Computing

PATENT

- **A privacy-preserving and anti-spoofing facial authentication system based on RFID (202110138649.X)**
Weiye Xu, Jianwei Liu, Jinsong Han, Feng Lin, le Cheng, Kui Ren
- **An anti-spoofing and mask-supported facial authentication system based on mmWave (202210609743.3)**
Weiye Xu, Wenfan Song, Jianwei Liu Ziyue Zhan, Yuanqing Zheng, Jinsong Han, Kui Ren

SELECTED HONORS AND AWARDS

- **Excellent Student Honor** 2019-2023
Zhejiang University
- **Xiaomi Scholarship (Top 1%)** 2021.10
Zhejiang University
- **Huawei Scholarship (Top 1%)** 2021.10
Zhejiang University
- **The Academic Star (Top 5%)** 2021.6
Zhejiang University
- **Best Paper Candidate in IEEE INFOCOM (7/251)** 2021.5
IEEE INFOCOM
- **National Scholarship (Top 1%)** 2020.10
The Ministry of education of China
- **Outstanding Graduate of School of Computer Science** 2019.6
Wuhan University
- **The Champion of the National College Student Internet of Things Contest** 2018.8

EXTERNAL REVIEWER

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT 2023)

INVITED TALKS

- **Mobicom 2022** 2022.10
Presenting our mmFace paper Virtually
- **Infocom 2021** 2021.5
Presenting our RFace paper Virtually
- **Mobicom 2020** 2020.9
Presenting our Deaf-aid paper Virtually

TEACHING ASSISTANT

- **Computer Network** Fall 2020, Fall 2021